20. Number Sequences (Set B)

Year 4 · Mark Scheme — Supplementary Worksheet

Foundation — Answers

1. 17

2. 48

3. C — 64

4. A, C and E (3,8,13,18 and 2,7,12,17 and 11,16,21,26 each go up by exactly 5; 6,10,14,18 goes up by 4, and 4,9,13,18 is not a steady gap)

5. 45

6. B — 55

7. 26

8. 24

9. B — 1250

10. A, C and D (each term is ×3, so the next term is 135×3=405, and the sequence is geometric; it is NOT arithmetic, since the gaps are 10, 30, 90, not constant)

11. 30

12. C — 42

13. 24 (9+15=24)

14. 40

15. A, B and D (4,9,16,25,36 are 2^2 to 6^2, so the next term is 7^2=49, and the gap grows each time: +5,+7,+9,+11; the gaps are not constant, so C is false, and the sequence increases, so E is false)

Standard — Answers

16. 41 (the gaps are +3,+5,+7,+9, so the next gap is +11)

17. 10

18. 37 (14+23=37)

19. 30

20. A, C and E (2,2,4,6,10 and 1,4,5,9,14 and 5,8,13,21,34 each follow the add-the-two-before-it rule; 3,6,9,12,15 goes up by 3 each time, and 2,4,8,16,32 doubles each time)

21. 49 (the gaps are +3,+6,+9,+12, so the next gap is +15)

22. 2, 7, 17, 37

23. 45 (triangular numbers add one more each time)

24. The rule is add an increasing odd gap each time (+3, +5, +7, +9…); the next term is 26+11=37

25. A, B, C and E (the gaps in A increase by 1 each time, B's gaps are a steady 4, C's gaps increase by 1 each time, and E's gaps increase by 1 each time — all count as forming their own arithmetic pattern. D's gaps double each time, 3,6,12, so they are NOT arithmetic)

26. 30

27. 7, 11, 18

28. 1440

29. A, C and D (5,10,20,40 is ×2, 200,100,50,25 is ÷2, and 6,18,54,162 is ×3, all geometric; 4,8,12,16 goes up by 4 (arithmetic), and 2,5,8,11 goes up by 3 (arithmetic))

30. 18, 30

31. C — 33

32. 7,14,28,56 is NOT arithmetic because the gaps are 7, 14, 28, which are not the same amount each time; it is a GEOMETRIC sequence, since each term is multiplied by 2

33. 28

34. 45

35. 12 (7+5=12)

Hard — Answers

36. 2160 (3×2=6, 6×3=18, 18×4=72, 72×5=360, 360×6=2160)

37. B — 57 (the gaps are 1, 4, 9, 16 — perfect squares — so the next gap is 25, and 32+25=57)

38. A, C and E (4,7,11,18,29 and 1,5,6,11,17 and 3,4,7,11,18 each follow the add-the-two-before-it rule; 2,3,6,9,15 does not, since 2+3=5 not 6; 10,15,20,25,30 is arithmetic, not Fibonacci-style)

39. 12, 34, 78 ((1+5)×2=12, (12+5)×2=34, (34+5)×2=78)

40. 126 (62×2+2=126)

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